

Quantum Fourier Transform

Seminar 2

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The Fourier transform, and Fourier analysis in general, are ubiquitous mathematical tools in physics. This is expected because many physical phenomena are wave-like in nature; that is, they involve the study of waves in order to be understood. Some important cases in which Fourier analysis is applied in physics include:

- Quantum mechanics.
- Seismology.
- Electromagnetic theory.
- Mechanical vibrations.
- Processing of natural and artificial signals.
- Spectroscopy.
- Solution of differential equations.
- Quantum field theory.
- Quantum computing.

1 Classical Fourier Transform

1.1 Fourier Series

A Fourier series can be defined as an expansion or representation of a function in a series of sines and cosines of the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx. \quad (1)$$

The conditions imposed on $f(x)$ for the previous equation to be valid are that $f(x)$ must have a finite number of finite discontinuities and a finite number of extrema, maxima and minima. These conditions are known as the Dirichlet conditions. Sturm–Liouville theory (see [1]) guarantees the validity of Eq. 1. Using orthogonality relations, the expansion coefficients can be obtained as

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos nt \, dt$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin nt \, dt, \quad n = 0, 1, 2, \dots$$

Substituting the coefficients a_n and b_n into Eq. 1, one obtains

$$\begin{aligned} f(x) &= \frac{1}{2\pi} \int_0^{2\pi} f(t) dt + \frac{1}{\pi} \sum_{n=1}^{\infty} \left[\cos nx \int_0^{2\pi} f(t) \cos nt \, dt + \sin nx \int_0^{2\pi} f(t) \sin nt \, dt \right] \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(t) dt + \frac{1}{\pi} \sum_{n=1}^{\infty} \int_0^{2\pi} f(t) \cos n(x-t) \, dt. \end{aligned} \tag{2}$$

The previous expressions are valid to represent the function $f(x)$ on the interval $[0, 2\pi]$. To represent it on an interval of length $2L$, Eq. 1 and its coefficients can be written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right] \tag{3}$$

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} \, dt \\ b_n &= \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} \, dt, \quad n = 0, 1, 2, \dots \end{aligned} \tag{4}$$

1.2 Fourier Integral

As mentioned above, Fourier series are especially useful for representing certain types of functions over a defined range: $[0, 2\pi]$, $[-L, L]$, and so on as needed, or over the interval $[-\infty, \infty]$ if the function is periodic. We now analyze the problem of representing a non-periodic function over an infinite range. Physically, this can be interpreted as the decomposition of an isolated pulse or wave packet into its sinusoidal components. The Fourier series coefficients on the interval $[-L, L]$, given by Eq. 4, lead to the Fourier series

$$\begin{aligned} f(x) &= \frac{1}{2L} \int_{-L}^L f(t) \, dt + \frac{1}{L} \sum_{n=1}^{\infty} \cos \frac{n\pi x}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} \, dt + \\ &\quad \frac{1}{L} \sum_{n=1}^{\infty} \sin \frac{n\pi x}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} \, dt. \end{aligned} \tag{5}$$

This can be rewritten as

$$f(x) = \frac{1}{2L} \int_{-L}^L f(t) \, dt + \frac{1}{L} \sum_{n=1}^{\infty} \int_{-L}^L f(t) \cos \frac{n\pi}{L} (t-x) \, dt. \tag{6}$$

Now let the parameter L approach infinity, transforming the finite interval $[-L, L]$ into the infinite interval $(-\infty, \infty)$. Define

$$\omega = \frac{n\pi}{L}, \quad \Delta\omega = \frac{\pi}{L}, \quad \text{with } L \rightarrow \infty. \tag{7}$$

With this, one obtains

$$f(x) \rightarrow \frac{1}{\pi} \sum_{n=1}^{\infty} \Delta\omega \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt, \quad (8)$$

and replacing the infinite sum by an integral over ω gives

$$f(x) = \frac{1}{\pi} \int_0^{\infty} d\omega \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt. \quad (9)$$

The a_0 term vanishes assuming that $\int_{-\infty}^{\infty} f(t) dt$ exists.

The result in Eq. 9 is known as the Fourier integral. It can be conveniently expressed in exponential form by noting that

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt, \quad (10)$$

and that

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega \int_{-\infty}^{\infty} f(t) \sin \omega(t-x) dt = 0. \quad (11)$$

These results follow from the fact that $\cos \omega(t-x)$ is an even function of ω , while $\sin \omega(t-x)$ is an odd function of ω . Adding Eq. 10 and Eq. 11 multiplied by the imaginary unit i , one obtains

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} d\omega \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt. \quad (12)$$

The variable ω is an arbitrary variable, also known as a dummy variable. It plays the same role as an index in a summation. However, in many physical problems, ω represents an angular frequency. Thus, Eqs. 10 and 12 can be interpreted as representations of $f(x)$ in terms of a distribution of infinite wavefronts with angular frequency ω , where the frequency is a continuous real variable.

1.3 Fourier Transform

Define a function $g(\omega)$ as the Fourier transform through

$$g(\omega) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt. \quad (13)$$

Then it is easy to see from the Fourier integral, Eq. 12, that

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{-i\omega x} d\omega. \quad (14)$$

This relation is also known as the inversion theorem or inverse Fourier transform. It is worth noting that Eqs. 13 and 14 are almost symmetric, but they differ in the sign of the imaginary unit i .

2 Quantum Fourier Transform

2.1 Introduction

As mentioned before, the Fourier transform has many applications throughout physics and also has important applications in classical computing, such as signal processing, data compression algorithms, and algorithmic complexity theory. The Quantum Fourier Transform is the quantum implementation of the discrete Fourier transform over the amplitudes of a wave function¹. It is a fundamental component of several quantum algorithms, such as Shor's factoring algorithm and quantum phase estimation.

The quantum Fourier transform acts on a vector $(x_0, \dots, x_{N-1})$ and maps it to another vector $(y_0, \dots, y_{N-1})$ through the formula

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \omega_N^{jk} \quad (15)$$

where

$$\omega_N^{jk} = e^{2\pi i \frac{jk}{N}}. \quad (16)$$

Analogously, the quantum Fourier transform acts on a quantum state²

$$|X\rangle = \sum_{j=0}^{N-1} x_j |j\rangle \quad (17)$$

and maps it to the state

$$|Y\rangle = \sum_{k=0}^{N-1} y_k |k\rangle \quad (18)$$

according to the expression

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \omega_N^{jk} \quad (19)$$

where ω_N^{jk} is defined in Eq. 16. It is important to note that only the amplitudes of the state are affected by this transformation. This can also be expressed as the mapping

$$|j\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{jk} |k\rangle \quad (20)$$

or by the unitary matrix

$$U_{\text{QFT}} = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} \omega_N^{jk} |k\rangle \langle j|. \quad (21)$$

¹A wave function that evolves according to the Schrödinger equation.

²A vector in a finite- or infinite-dimensional Hilbert space.

2.1.1 Intuition Behind the QFT

The intuition behind the Quantum Fourier Transform (QFT) comes from the fact that, when it transforms a quantum state, it essentially changes from the computational Z basis to the Fourier basis. The Hadamard gate ([4], [9], [6]) is the quantum Fourier transform for a single qubit: it transforms from the computational Z basis $\{|0\rangle, |1\rangle\}$ to the X basis $\{|+\rangle, |-\rangle\}$. In the same way, every multi-particle state, that is, every multi-qubit state in the computational basis, has a representation in the Fourier basis. The QFT is simply the function that transforms between these bases.

$$\text{QFT}|x\rangle = |\tilde{x}\rangle, \quad (22)$$

where the Fourier basis is denoted by $|\tilde{x}\rangle$.

2.2 Example 1: One-Qubit QFT

We begin by considering how the previously defined QFT operator acts on a single-qubit state, i.e. $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. For this case, $x_0 = \alpha$, $x_1 = \beta$, and $N = 2$. Therefore, the first Fourier coefficient in the transformation is

$$y_0 = \frac{1}{\sqrt{2}} \left(\alpha \exp\left(2\pi i \frac{0 \times 0}{2}\right) + \beta \exp\left(2\pi i \frac{1 \times 0}{2}\right) \right) = \frac{1}{\sqrt{2}}(\alpha + \beta) \quad (23)$$

and for the second coefficient,

$$y_1 = \frac{1}{\sqrt{2}} \left(\alpha \exp\left(2\pi i \frac{0 \times 1}{2}\right) + \beta \exp\left(2\pi i \frac{1 \times 1}{2}\right) \right) = \frac{1}{\sqrt{2}}(\alpha - \beta). \quad (24)$$

Thus, the total result on $|\psi\rangle$ is

$$U_{\text{QFT}}|\psi\rangle = \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle. \quad (25)$$

This operation is equivalent to applying the Hadamard operator

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (26)$$

to the state $|\psi\rangle$,

$$\begin{aligned} H|\psi\rangle &= \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle \equiv \tilde{\alpha}|0\rangle + \tilde{\beta}|1\rangle \\ &= \alpha|+\rangle + \beta|-\rangle. \end{aligned} \quad (27)$$

2.3 Quantum Fourier Transform: General Definition

The previous example does not make completely clear how a quantum Fourier transform is applied for larger N . Here, take $N = 2^n$, that is, U_{QFT_N} acts on the state $|x\rangle = |x_1 \dots x_n\rangle^3$, where x_1 is the most significant bit. Applying Eq.

³We use the notation $|x_1 \dots x_n\rangle = |x_1\rangle \otimes \dots \otimes |x_n\rangle$.

18, one obtains

$$\begin{aligned}
U_{\text{QFT}_N}|x\rangle &= \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega_N^{xy} |y\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{2\pi i xy/2^n} |y\rangle.
\end{aligned} \tag{28}$$

Rewriting y in binary fractional notation $y = y_1 \dots y_n$ with

$$\frac{y}{2^n} = \sum_{k=1}^n \frac{y_k}{2^k},$$

where $y_k \in \{0, 1\}$, and substituting into Eq. 28, one obtains

$$\begin{aligned}
U_{\text{QFT}_N}|x\rangle &= \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{2\pi i x (\sum_{k=1}^n y_k / 2^k)} |y_1 \dots y_n\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \prod_{k=1}^n e^{2\pi i x y_k / 2^k} |y_1 \dots y_n\rangle.
\end{aligned} \tag{29}$$

Here, the exponential of a sum has been expressed as a product of exponentials. Rearranging the sums and products, and expanding $\sum_{y=0}^{N-1} = \sum_{y_1=0}^1 \sum_{y_2=0}^1 \dots \sum_{y_n=0}^1$, gives

$$\begin{aligned}
U_{\text{QFT}_N}|x\rangle &= \frac{1}{\sqrt{N}} \bigotimes_{k=1}^n \left(|0\rangle + e^{2\pi i x / 2^k} |1\rangle \right) \\
&= \frac{1}{\sqrt{N}} \left(|0\rangle + e^{\frac{2\pi i}{2} x} |1\rangle \right) \otimes \left(|0\rangle + e^{\frac{2\pi i}{2^2} x} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\frac{2\pi i}{2^n} x} |1\rangle \right).
\end{aligned} \tag{30}$$

3 Circuit for the Quantum Fourier Transform

The quantum Fourier transform can be implemented using the quantum circuit model used by several real implementations of quantum computing. For an introduction to these circuits, see [9], [4], [6], etc. To build the QFT circuit, two gates are used. The first is a single-qubit gate, the Hadamard gate in Eq. 26. From the discussion of the one-qubit QFT example and from the derivation of the general expression for the QFT, it is known that, for the single-qubit state $|x_k\rangle$, the corresponding quantum Fourier transform is

$$H|x_k\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + \exp\left(\frac{2\pi i}{2} x_k\right) |1\rangle \right). \tag{31}$$

The second two-qubit gate is the controlled rotation $C\text{ROT}_k$, which in block-matrix notation is defined as

$$C\text{ROT}_k = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & U\text{ROT}_k \end{bmatrix} \tag{32}$$

where

$$U \text{ROT}_k = \begin{bmatrix} 1 & 0 \\ 0 & \exp\left(\frac{2\pi i}{2^k}\right) \end{bmatrix}. \quad (33)$$

The effect of applying $C \text{ROT}_k$ to a two-qubit state $|x_l x_j\rangle$, where the first qubit is the control qubit and the second qubit is the target, is given by

$$C \text{ROT}_k |0 x_j\rangle = |0 x_j\rangle \quad (34)$$

and

$$C \text{ROT}_k |1 x_j\rangle = \exp\left(\frac{2\pi i}{2^k} x_j\right) |1 x_j\rangle. \quad (35)$$

With these two gates, the circuit that implements the quantum Fourier transform can be built as shown in the following diagram.

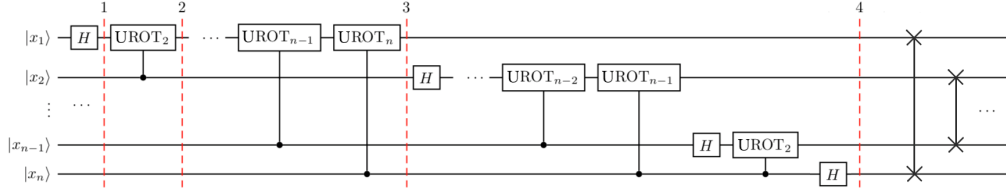


Figure 1: QFT circuit implementation.

The circuit operates as follows. We start with an input state $|x_1 x_2 \dots x_n\rangle$.

1. Apply the Hadamard gate to the first qubit, transforming the input state as follows:

$$H_1 |x_1 x_2 \dots x_n\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2} x_1\right) |1\rangle \right] \otimes |x_2 x_3 \dots x_n\rangle. \quad (36)$$

2. Then apply the $U \text{ROT}_2$ gate to qubit 1 controlled by qubit 2. This results in

$$\frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^2} x_2 + \frac{2\pi i}{2} x_1\right) |1\rangle \right] \otimes |x_2 x_3 \dots x_n\rangle. \quad (37)$$

3. This process is repeated until qubit n is reached. That is, when the last gate $U \text{ROT}_n$ is applied to qubit 1 controlled by qubit n , the state is transformed into

$$\frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^n} x_n + \frac{2\pi i}{2^{n-1}} x_{n-1} + \dots + \frac{2\pi i}{2^2} x_2 + \frac{2\pi i}{2} x_1\right) |1\rangle \right] \otimes |x_2 x_3 \dots x_n\rangle. \quad (38)$$

Noting that⁴

$$x = 2^{n-1} x_1 + 2^{n-2} x_2 + \dots + 2^1 x_{n-1} + 2^0 x_n, \quad (39)$$

the state in Eq. 38 can be written as

$$\frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^n} x\right) |1\rangle \right] \otimes |x_2 x_3 \dots x_n\rangle. \quad (40)$$

⁴At the end, x is a label for the state and $\{x_1, \dots, x_n\} \in \{0, 1\}$, so this is a binary, i.e. base-2, representation of the label.

4. After an analogous sequence to steps 1–3 for qubits $2 \dots n$, the final state is

$$\begin{aligned} & \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^n} x\right) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^{n-1}} x\right) |1\rangle \right] \otimes \dots \\ & \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^2} x\right) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^1} x\right) |1\rangle \right]. \end{aligned} \quad (41)$$

This is equivalent to the general expression in Eq. 30 developed previously, with the observation that the order of the qubits has been reversed in the final output state. For this reason, operators called *swap* gates are used, as shown at the end of the diagram in Fig. 1.

3.1 Example 2: Three-Qubit QFT

To make the calculations explicit for the three-qubit QFT, we want to find $|y_1 y_2 y_3\rangle = QFT_8 |x_3 x_2 x_1\rangle$. The steps are the following:

1. Apply the Hadamard gate to qubit $|x_1\rangle$:

$$|\psi_1\rangle = |x_3\rangle \otimes |x_2\rangle \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2} x_1\right) |1\rangle \right]. \quad (42)$$

2. Apply the $U \text{ROT}_2$ gate to qubit $|x_1\rangle$, controlled by qubit $|x_2\rangle$:

$$|\psi_2\rangle = |x_3\rangle \otimes |x_2\rangle \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^2} x_2 + \frac{2\pi i}{2} x_1\right) |1\rangle \right]. \quad (43)$$

3. Apply the $U \text{ROT}_3$ gate to $|x_1\rangle$, controlled by $|x_3\rangle$:

$$|\psi_3\rangle = |x_3\rangle \otimes |x_2\rangle \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^3} x_3 + \frac{2\pi i}{2^2} x_2 + \frac{2\pi i}{2} x_1\right) |1\rangle \right]. \quad (44)$$

4. Now begin operating on the second qubit⁵, again with the Hadamard gate:

$$|\psi_4\rangle = |x_3\rangle \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2} x_2\right) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^3} x_3 + \frac{2\pi i}{2^2} x_2 + \frac{2\pi i}{2} x_1\right) |1\rangle \right]. \quad (45)$$

5. Apply the $U \text{ROT}_2$ gate to qubit $|x_2\rangle$, controlled by qubit $|x_3\rangle$:

$$|\psi_5\rangle = |x_3\rangle \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^2} x_3 + \frac{2\pi i}{2} x_2\right) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^3} x_3 + \frac{2\pi i}{2^2} x_2 + \frac{2\pi i}{2} x_1\right) |1\rangle \right]. \quad (46)$$

6. Only the Hadamard gate is applied to the third qubit $|x_3\rangle$:

$$\begin{aligned} |\psi_6\rangle = & \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2} x_3\right) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^2} x_3 + \frac{2\pi i}{2} x_2\right) |1\rangle \right] \\ & \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp\left(\frac{2\pi i}{2^3} x_3 + \frac{2\pi i}{2^2} x_2 + \frac{2\pi i}{2} x_1\right) |1\rangle \right]. \end{aligned} \quad (47)$$

⁵This would generally be obtained mathematically by acting on the tensor-product state with the tensor product of operators for each qubit, but in the circuit approach these calculations are performed sequentially.

7. Finally, note that the result does not have the desired order for the QFT defined previously. Therefore, the order of the qubits must be reversed; in this particular case, qubit 3 must be swapped with qubit 1.

3.2 Remarks on the QFT Circuit

In the previous three-qubit transform example, a very useful form of the QFT for $N = 2^n$ is demonstrated. In Eq. 47, only the last qubit depends on the value of all the input qubits, and each preceding qubit has progressively less dependence on the input qubits. This becomes relevant in physical implementations of this circuit, where couplings between nearby qubits are easier to obtain than couplings between distant qubits.

Additionally, as the circuit for the QFT implementation becomes larger, more time is spent performing incrementally small rotations. It turns out that, after a certain threshold, these rotations can be ignored while still obtaining an acceptable result. This is also very important for physical implementations because reducing the number of operations can considerably decrease decoherence and the intrinsic errors associated with applying many quantum gates.

3.3 Implementation in Qiskit

Qiskit is the platform created by IBM to run quantum circuits on its quantum computers. It provides options to simulate circuits and options to run them on quantum computers distributed around the world. Qiskit is relatively easy to use and is available as an API in the popular open-source language Python.

In Qiskit, the implementation of the $CROT$ gate used in the previous derivations is known as the controlled phase rotation. This gate is defined in **OpenQASM** [10] as

$$CP(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\theta} \end{bmatrix}. \quad (48)$$

Therefore, to obtain the $CROT_k$ gate through the CP gate, the mapping is

$$\theta = \frac{2\pi}{2^k} = \frac{\pi}{2^{k-1}}. \quad (49)$$

3.3.1 Three-Qubit QFT

Using the Qiskit quantum computing platform developed by IBM, the QFT can be implemented through a Python API. The resulting circuit diagram, also generated with Qiskit tools, can be seen in the following figure.

3.3.2 QFT for n Qubits

To apply the general transform to n qubits, the previous approach is used together with a programming technique: recursion. The gates are applied to the most significant qubit, and then the same function is called recursively to apply the necessary gates to the next most significant qubit, and so on until all qubits in the circuit have been operated on. To see the sequential circuit diagram, the function was called for 6 qubits. The result is the following.

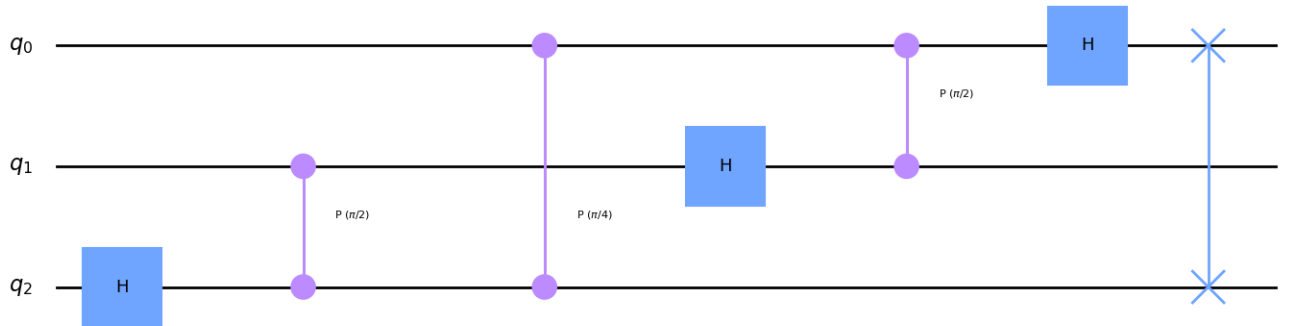


Figure 2: QFT circuit implementation for three qubits.

The previous implementations can be found in the GitHub repository: <https://github.com/Julio-Medina/Quantum-Fourier-Transform>

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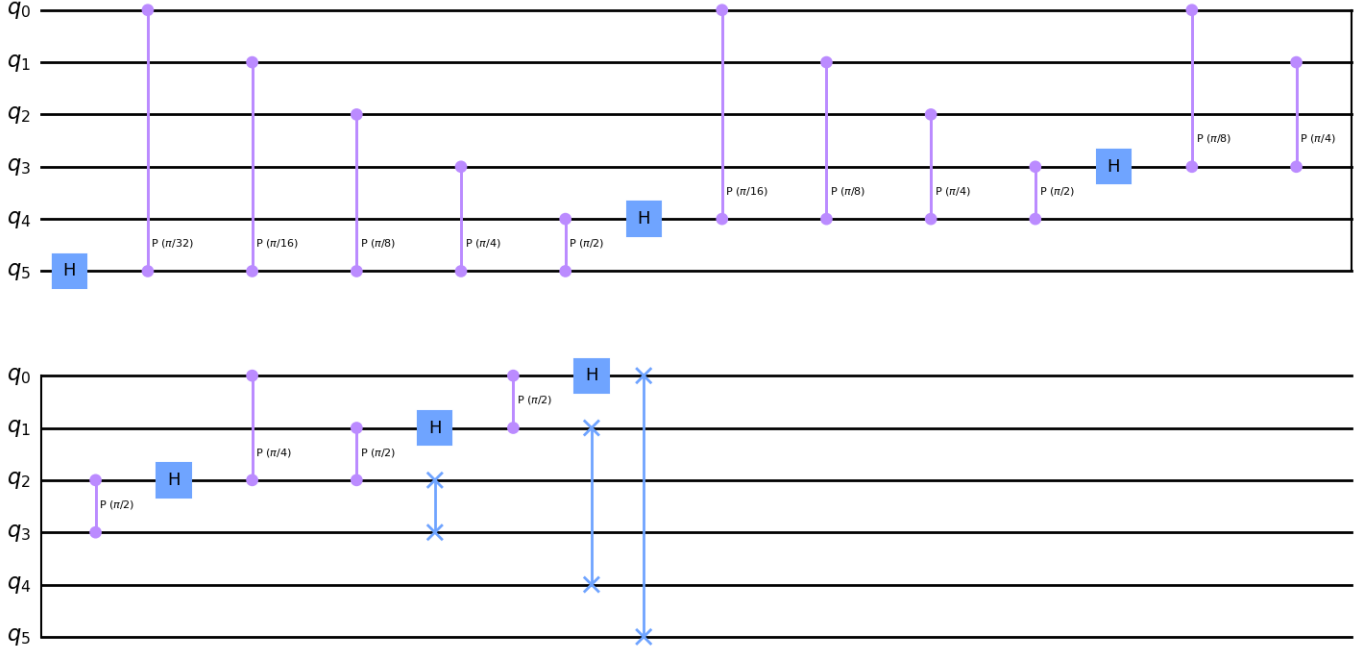


Figure 3: QFT circuit implementation for six qubits.

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